Murray Patterson

Curriculum Vitae

Education

2006—2012 **Doctor of Philosophy**, *Department of Computer Science, University of British Columbia*, Vancouver, Canada, *GPA – 4.0/4*.
Advisors: Ján Maňuch, Cedric Chauve and Arvind Gupta
Thesis title: Variants of the consecutive-ones property motivated by the reconstruction of ancestral species

2004—2006 **Master of Science**, *School of Computing Science, Simon Fraser University*, Burnaby, British Columbia, Canada, *GPA – 3.9/4*.
Advisors: Eugenia Ternovska and Arvind Gupta
Thesis title: Grounding for model expansion in k-guarded formulas

1999—2003 **Bachelor of Computer Science (Honours)**, *Jodrey School of Computer Science, Acadia University*, Wolfville, Nova Scotia, Canada, *GPA – 3.8/4 (top 5%)*.
Advisors: Raymond J. Spiteri and Jim Diamond
Thesis title: Implementing Runge-Kutta solvers in Java

Work Experience

2020— **Assistant Professor**, *Department of Computer Science, Georgia State University*, Atlanta, Georgia, USA.
Teaching both undergraduate and graduate courses in computer science, carrying out a research program in bioinformatics, and serving the department, and the university

2019—2020 **Visiting Assistant Professor**, *Department of Computer Science and Engineering, Fairfield University*, Fairfield, Connecticut, USA.
Teaching both undergraduate and graduate courses in computer science, as well as carrying out my research program in bioinformatics

2016—2019 **Postdoctoral Researcher**, *Experimental Algorithmics Lab (AlgoLab), Informatics Department (DISCo), Università degli Studi di Milano-Bicocca*, Milano, Italy.
Advisors: Gianluca Della Vedova and Paola Bonizzoni
Performing research on haplotype assembly, cancer phylogenetics and their application

2014—2016 **Postdoctoral Researcher**, *Laboratoire de Biométrie et Biologie Évolutive (LBBE), Université Claude Bernard Lyon 1*, Villeurbanne, France.
Advisors: Daniel Kahn and Vincent Daubin
Studying the co-evolution of (metabolic) function, and how it can be used to help the construction of phylogenetic trees

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2013—2014 **ERCIM Postdoctoral Research Fellow**, *Life Sciences Group, Centrum Wiskunde Informatica (CWI)*, Amsterdam, The Netherlands.
Advisors: Alexander Schönhuth and Gunnar W. Klau
Improving the state-of-the-art haplotype assembly methods for next-generation genome sequencing data

2012—2013 **ERCIM Postdoctoral Research Fellow**, *INRIA Rhône-Alpes and the Evolutionary Biology Lab (LBBE), Université Claude Bernard Lyon 1*, Villeurbanne, France.
Advisor: Eric Tannier
Performing a study of the co-evolution of genes, and how it can be used to help the construction of phylogenetic trees

Teaching Experience

2020— **Department of Computer Science**, *Georgia State University*, Atlanta, Georgia, USA.

Spring 2026 CSC 4330 / 6330 — a senior year / masters-level programming language concepts course (two sections)

Fall 2025 CSC 4510 / 6510 — a senior year / masters-level course on automata theory

Fall 2024 CSC 4520 / 6520 — a senior / masters course on algorithm design / analysis; and
CSC 8540 — a grad-level course on advanced algorithms in bioinformatics

Spring 2024 CSC 4330 / 6330 — a senior / masters programming language concepts course

Fall 2023 CSC 4330 / 6330 — a senior / masters programming language concepts course

Spring 2023 CSC / DSCI 4810 / 6810 — a senior year / master-level course on artificial intelligence

Fall 2022 CSC 4330 / 6330 — a senior / masters programming language concepts course

Spring 2022 CSC 4330 / 6330 — a senior / masters programming language concepts course

Fall 2021 CSC 4710 / 6710 — a senior year / masters-level course on database systems

Spring 2021 CSC 4330 / 6330 — a senior / masters programming language concepts course

Fall 2020 CSC 4330 / 6330 — a senior / masters programming language concepts course

2019—2020 **Department of Computer Science and Engineering**, *Fairfield University*, Fairfield, Connecticut, USA.

Spring 2020 CS 131 — an introductory programming course in Java; and
CS 354 — a junior year theory of programming languages course

Fall 2019 CS 131 — an introductory programming course in Java;
CS 231 — an intermediate programming course, focusing on object-oriented Java; and
SW 402 — a senior-year / masters-level course on database management systems

2015 **Department of Mathematics**, *Université Claude Bernard*, Villeurbanne, France.
Section head: Raymond Lobry
Teaching a section of mathematics for the life sciences to first-year university students

Advising Students

Spring 2026 I supervised PhD students Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed, and Master's student Ramita Rathore

- Ramita Rathore successfully completed her Master's project titled: "An efficient algorithm for updating APSP under edge removal".

Fall 2025 I supervised PhD students Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed, and Master's student of the Biology Department James Dang

- Mansoor Ahmed won an MBD fellowship, funding the remainder of his study.
- Prakash Chourasia received the 2025 MBD Outstanding Fellow Award.
- James Dang successfully completed his Master's project titled: "Analysis of the chromatophores involved in anemonefish coloration using bioinformatics to identify DUOX1 and other genes in the chromatophores".

Spring 2025 I supervised PhD students Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed

- Prakash Chourasia received the 2025 Outstanding Graduate Student Award in the Department of Computer Science.

Fall 2024 I supervised PhD students Sarwan Ali, Prakash Chourasia, Avais Jan, Ayotomiwa Ezekiel Adeniyi and Mansoor Ahmed, Master's student Sai Roopesh Mandava, and Undergraduate student Ramita Rathore

- Sai Roopesh Mandava successfully completed his Master's project titled: "Kickstarting football skills: A reinforcement learning approach".
- Sarwan Ali successfully defended his PhD dissertation titled: "Designing representation learning methods for molecular sequences analysis".
- Sarwan Ali received a Georgia State dissertation grant award.

Spring 2024 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia, Taslim Murad, Avais Jan and Ayotomiwa Ezekiel Adeniyi, Master's students Raghuveer Gummadi and Sai Roopesh Mandava, and Undergraduate students Ramita Rathore and Anderson Hernandez

- Taslim Murad successfully defended her PhD dissertation titled: "Designing methods for representation learning of molecular sequences and its application in analysis tasks".
- Zahra Tayebi successfully defended her PhD dissertation titled: "Clustering, classification, and simulation of evolutionarily related sequences in the context of cancer and viral spread".
- Sarwan Ali received the 2024 Graduate Research Award in the Department of Computer Science.
- Ramita Rathore won a 1st Place Novice Research Award at the 2024 GSU Summer Undergraduate Research Symposium.

Fall 2023 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia, Taslim Murad, Avais Jain and Ayotomiwa Ezekiel Adeniyi, and Master's students Raghuveer Gummadi and Eunice Olorunshola.

- Prakash Chourasia won an MBD fellowship, funding the remainder of his study.
- Zahra Tayebi successfully completed her dissertation proposal titled: "Efficient clustering and classification of protein sequences".
- Sarwan Ali successfully completed his dissertation proposal titled: "Designing representation learning methods for molecular sequences analysis".
- Eunice Olorunshola successfully defended her Master's thesis titled: "Classifying different cancer types based on transcriptomics data using machine learning algorithms".

Spring 2023 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia, Taslim Murad and Avais Jain, and Master's students Raghuveer Gummadi, Eunice Olorunshola, Sai Venkata Sumanth Gaddam, Kushal Khatri, Surya Subrahmanya Sai Chandra Sekhar Kolli, Samvidha Reddy Mannem, Fatma Noor, Ashwin Pilgaonkar, Ria C. Thazhe Punathil, Sainikhila Vemulapalli and Karishma Mahendra Wadhe.

- Taslim Murad successfully completed her dissertation proposal titled: "Designing methods for representation learning of biological sequences and its application in analysis tasks".

The following Master's projects were graduated:

- Sai Gaddam: "Diabetic retinopathy detection using deep learning".
- Kushal Khatri: "A stratified subsampler for hierarchically structured data".
- Surya Kolli: "Recommendations using graph databases".
- Fatma Noor: "Can community detection build a good phylogeny?".
- Ashwin Pilgaonkar: "Robinson Country Intelligence Index (RCII) – A country risk analysis tool".
- Ria Punathil: "CoVScan – A tool for viral host specification from coronavirus spike sequences".
- Sainikhila Vemulapalli: "Vehicle image classification and damage analysis using deep learning model".
- Karishma Wadhe: "Drive safe – A driver drowsiness alert system".

Fall 2022 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia and Taslim Murad, and Master's students Supriya Arun, Kaustubh Khandai, Kushal Khatri, Prashanth Annam, Babitha Reddy Devireddy, Indu Munnangi, Venkata Krishna Chaitanya Nuthalapati, Dhruval Patel, Shraddha Sharma and Atharv Utkarsh Yeolekar.

- Zahra Tayebi won a B&B fellowship, funding the remainder of her study.

The following Master's projects were graduated:

- Supriya Arun: "Comparative analysis of Covid classifiers".
- Kaustubh Khandai: "Determining significant correlation between pairs of extant characters in a small parsimony framework".

- Prashanth Annam: "Neobook: A social medial application using Flutter and Firebase".
- Babitha Reddy Devireddy: "Chaos engineering using Gremlin and GC analyzer".
- Indu Munnangi: "My recipe book: A MERN stack web application".
- Venkata Krishna Chaitanya Nuthalapati: "Tsuba: Web security scanner".
- Dhruval Patel: "Best actor detection".
- Shraddha Sharma: "Sunsetting customer identification service".
- Atharv Utkarsh Yeolekar: "Data condensation using generative adversarial networks".

Spring 2022 I supervised PhD students Zahra Tayebi, Sarwan Ali, Prakash Chourasia and Taslim Murad, and Master's students Babatunde Bello, Supriya Arun and Kaustubh Khandai.

- Babatunde Bello graduated with Master's thesis titled: "Functional enrichment analysis of transcriptomics data of breast cancer RNA-seq".

Fall 2021 PhD students Zahra Tayebi, Sarwan Ali and Prakash Chourasia, Master's students Babatunde Bello and Kaustubh Khandai, and Honors student Yijing Zhou.

- Sarwan Ali won an MBD fellowship, funding the remainder of his study.
- Yijing Zhou graduated with honors thesis titled: "Towards a complete workflow for inferring tumor phylogeny from raw single-cell sequencing reads"

Spring 2021 I supervised PhD students Zahra Tayebi and Sarwan Ali

Fall 2020 I supervised PhD student Zahra Tayebi

Volunteer Experience

2017—2019 **Research Advisor**, *Lycée Stendhal*, Milan, Italy.
Advising and guiding students on an after-school mathematics project which is part of the larger European program: Math en Jeans. There is an annual conference where the students go to present their work on the year's project.

Grants

Pending

2026 **AIDatasets: Impact: Reconciling multi-source, multi-resolution data for AI-ready energy analysis**, *NSF*, Amount: $669,961 (3 years).
PI: Chetan Tiwari, Co-I: Murray Patterson, Armin R. Mikler

2026 **Learning evolutionary patterns from longitudinal viral and tumor sequencing data**, *NSF*, Amount: $599,727 (3 years).
PI: Murray Patterson, Co-I: Alexander Zelikovsky

2026 **Effective hazard mitigation through regional risk ranking**, *Nokia Corporation*, Amount: 98,200 € (2 years).
PI: Murray Patterson, Co-I: Armin R .Mikler

Funded

2024 **Using AI techniques to uncover transmission dynamics and hotspots from HIV big data**, *Research Internal Grant (RIG), GSU*, Amount: $20,000 USD (1 year).
PI: Alexander Kirpich, Co-I: Murray Patterson

2024 **REU site: Multidisciplinary graph data analytics**, *National Science Foundation*, Amount: $372,720 USD (3 years).
PI: Esra Akbas, Mehmet Emin Aktas, Mentor: Murray Patterson

2021 **AI Learning Alliance: Broadening participation**, *Meta, Inc.*, Amount: $50,000 USD (2 years).
PI: Anu Bourgeois, Co-I: Murray Patterson

2021 **Global education online module design**, *Hanyang University ERICA Industry-University Cooperation Foundation*, Amount: $24,255 USD (3 years).
PI: Zhipeng Cai, Co-I: Armin R. Mikler, Rajshekhar Sunderraman, Pavel Skums, Murray Patterson, Rolando Estrada, Yingshu Li, Jennifer Sherer, Geoffrey Graybeal

Awards and Honors

2023 Nomination for membership to Sigma Xi, the scientific research honor society

2011 The Marie Curie Alain Bensoussan (ABCDE) Postdoctoral Fellowship of the European Research Consortium for Informatics and Mathematics (ERCIM)
Award of 60,300 € and 5000 € for travel, in order to fully fund two years of postdoctoral study in the European ERCIM member countries

2007 The Postgraduate Scholarship (PGS) of the Natural Sciences and Engineering Research Council (NSERC) of Canada
Award of $75,116 CAD in order to fully fund three years of postgraduate (PhD) research

2006 Simon Fraser University Computing Science Graduate Fellowship
Award of $3000 CAD towards funding postgraduate research

2005 Simon Fraser University Computing Science Graduate Fellowship
Award of $3000 CAD towards funding postgraduate research

2001 The Undergraduate Student Research Award (USRA) of the Natural Sciences and Engineering Research Council (NSERC) of Canada
Award of $8500 CAD in order to fully fund two summers of undergraduate research

Technical Skills

advanced	C/C++, Python, LaTeX	*10+ years experience*
intermediate	Java, Unix tools, Shell programming, Prolog, Haskell	*5+ years experience*
basic	Matlab, R	*3+ years experience*

Languages

English	maternal language	
French	second language	*conversational and technical fluency*
Italian	intermediate	*comprehension and basic conversational skills*
Spanish	intermediate	*comprehension and basic conversational skills*
中文	basic	*basic words and phrases*

Publications

Journal Articles

1. Jingbo Yang and Murray Patterson. Alignment-free k-mer versus species-abundance profiling for metagenomic classification: A comparative cross-cohort evaluation. *Medical & Biological Engineering & Computing*, 2026. to appear.
2. Mansoor Ahmed, Nadeem Taj, Imdad Ullah Khan, Hemanth Venkateswara, and Murray Patterson. CHIMERA-Bench: A benchmark dataset for epitope-specific antibody design. *Transactions on Machine Learning Research*, 2026. to appear. URL: `https://openreview.net/forum?id=3m8uaq7luq`.
3. Sarwan Ali, Prakash Chourasia, and Murray Patterson. Anderson acceleration for molecular sequencing. *Neural Computing and Applications*, 2026. to appear.
4. Sarwan Ali, Haris Mansoor, and Murray Patterson. Breaking the Euclidean barrier: Hyperboloid-based biological sequence analysis. *Neural Computing and Applications*, 38(18):746, 2026. doi:10.1007/s00521-026-12468-3.
5. Armin R. Mikler, Chetan Tiwari, and Murray Patterson. A spatial alignment problem. *MDPI Algorithms*, 19(6):475, 2026. doi:10.3390/a19060475.
6. Sarwan Ali, Tamkanat E. Ali, Prakash Chourasia, and Murray Patterson. Compression and k-mer based approach for anticancer peptide analysis. *Transactions on Computational Biology and Bioinformatics*, pages 1–12, 2026. doi:10.1109/TCBBIO.2026.3685197.
7. Taslim Murad, Prakash Chourasia, Sarwan Ali, Imdad Ullah Khan, and Murray Patterson. DANCE: Deep learning-assisted Analysis of proteiN sequences using Chaos Enhanced kaleidoscopic images. *Journal of Computational Biology*, 33(5-6):680–691, 2026. doi:10.1177/15578666261441311.
8. Ezekiel Ayotomiwa Adeniyi, Akshay Juyal, Pavel Skums, Murray Patterson, and Alex Zelikovsky. Uncovering epistatic interactions in SARS-CoV-2 evolution through hidden Markov models. *Journal of Computational Biology*, 33(5-6):692–704, 2026. doi:10.1177/15578666261423963.
9. Sarwan Ali, Prakash Chourasia, Bipin Koirala, and Murray Patterson. Nearest neighbor CCP-based molecular sequence analysis. *Transactions on Computational Biology and Bioinformatics*, 23(1):1–13, 2026. doi:10.1109/TCBBIO.2025.3621138.

10. Simone Ciccolella, Murray Patterson, Iman Hajirasouliha, and Gianluca Della Vedova. Cancer progression inference using a finite-state model to allow recurrences and losses of mutations. *Neural Computing and Applications*, 37:21545–21562, 2025. doi:10.1007/s00521-025-11474-1.
11. Sarwan Ali, Tamkanat E. Ali, and Murray Patterson. Hash-binding: DNA-protein binding prediction using hashing-based embedding. *Neural Computing and Applications*, 37:16905–16940, 2025. doi:10.1007/s00521-025-11349-5.
12. Sarwan Ali, Tamkanat E. Ali, Haris Mansoor, Prakash Chourasia, and Murray Patterson. Hist2Vec: A histogram and kernel-based embedding method for molecular sequence analysis. *Expert Systems with Applications*, 273:126859, 2025. doi:10.1016/j.eswa.2025.126859.
13. Sarwan Ali, Madiha Shabbir, Haris Mansoor, Prakash Chourasia, and Murray Patterson. Elliptic geometry-based kernel matrix for improved biological sequence classification. *Knowledge-Based Systems*, 304:112479, 2024. doi:10.1016/j.knosys.2024.112479.
14. Zahra Tayebi, Sarwan Ali, and Murray Patterson. TCellR2Vec: Efficient feature selection for TCR sequences for cancer classification. *PeerJ Computer Science*, 10:e2239, 2024. doi:10.7717/peerj-cs.2239.
15. Sarwan Ali, Tamkanat E. Ali, Taslim Murad, Haris Mansoor, and Murray Patterson. Molecular sequence classification using efficient kernel based embedding. *Information Sciences*, 679:121100, 2024. doi:10.1016/j.ins.2024.121100.
16. Sarwan Ali, Prakash Chourasia, and Murray Patterson. From PDB files to protein features: A comparative analysis of PDB bind and STCRDAB datasets. *Medical & Biological Engineering & Computing*, 62:2449–2483, 2024. doi:10.1007/s11517-024-03074-3.
17. Zahra Tayebi, Sarwan Ali, Taslim Murad, Imdad Ullah Khan, and Murray Patterson. PseAAC2Vec protein encoding for TCR protein sequence classification. *Computers in Biology and Medicine*, 170:107956, 2024. doi:10.1016/j.compbiomed.2024.107956.
18. Sarwan Ali, Prakash Chourasia, and Murray Patterson. When protein structure embedding meets large language models. *MDPI Genes*, 15(1):25, 2023. doi:10.3390/genes15010025.
19. Sarwan Ali and Murray Patterson. Improving ISOMAP efficiency with RKS: A comparative study with t-distributed neighbor embedding on protein sequences. *MDPI J — Multidisciplinary Scientific Journal*, 6(4):579–591, 2023. doi:10.3390/j6040038.
20. Taslim Murad*, Sarwan Ali*, Imdad Ullah Khan, and Murray Patterson. Spike2CGR: An efficient method for spike sequence classification using chaos game representation. *Machine Learning*, 112:3633–3658, 2023. doi:10.1007/s10994-023-06371-4.
21. Sarwan Ali, Prakash Chourasia, Zahra Tayebi, Babatunde Bello, and Murray Patterson. ViralVectors: Compact and scalable alignment-free virome feature generation. *Medical & Biological Engineering & Computing*, 2023. doi:10.1007/s11517-023-02837-8.

22. Taslim Murad, Sarwan Ali, and Murray Patterson. Exploring the potential of GANs in biological sequence analysis. *MDPI Biology*, 12(6):854, 2023. doi:10.3390/biology12060854.

23. Bikram Sahoo, Sarwan Ali, Pin-Yu Chen, Murray Patterson, and Alexander Zelikovsky. Assessing the resilience of machine learning classification algorithms on SARS-CoV-2 genome sequences generated with long-read specific errors. *Biomolecules*, 13(6):934, 2023. doi:10.3390/biom13060934.

24. Danushka Bandara, Karen Exantus, Cristian Navarro-Martinez, Murray Patterson, and Ashley Byun. Identifying distinguishing acoustic features in felid vocalizations based on call type and species classification. *Acoustics Australia*, 2023. doi:10.1007/s40857-023-00298-5.

25. Sarwan Ali, Babatunde Bello, and Murray Patterson. Solvent accessibility of Coronaviridae spike proteins through the lens of information gain. *MDPI J — Multidisciplinary Scientific Journal*, 6(2):236–247, 2023. doi:10.3390/j6020018.

26. Sarwan Ali, Bikram Sahoo, Alexander Zelikovsky, Pin-Yu Chen, and Murray Patterson. Benchmarking machine learning robustness in Covid-19 genome sequence classification. *Scientific Reports*, 13(1):4154, 2023. doi:10.1038/s41598-023-31368-3.

27. Sarwan Ali, Babatunde Bello, Zahra Tayebi, and Murray Patterson. Characterizing SARS-CoV-2 spike sequences based on geographical location. *Journal of Computational Biology*, 30(4):432–445, 2023. doi:10.1089/cmb.2022.0391.

28. Prakash Chourasia, Sarwan Ali, Simone Ciccolella, Gianluca Della Vedova, and Murray Patterson. Reads2Vec: Efficient embedding of raw high-throughput sequencing reads data. *Journal of Computational Biology*, 30(4):469–491, 2023. doi:10.1089/cmb.2022.0424.

29. Sarwan Ali, Bikram Sahoo, Muhammad Asad Khan, Alexander Zelikovsky, Imdad Ullah Khan*, and Murray Patterson*. Efficient approximate kernel based spike sequence classification. *Transactions on Computational Biology and Bioinformatics*, 2023. doi:10.1109/TCBB.2022.3206284.

30. Kaustubh Khandai, Cristian Navarro-Martinez, Brendan Smith, Rebecca Buonopane, S. Ashley Byun*, and Murray Patterson*. Determining significant correlation between pairs of extant characters in a small parsimony framework. *Journal of Computational Biology*, 29(10):1132–1154, 2022. doi:10.1089/cmb.2022.0141.

31. Sarwan Ali, Babatunde Bello, Prakash Chourasia, Ria Thazhe Punathil, Yijing Zhou, and Murray Patterson. PWM2Vec: An efficient embedding approach for viral host specification from coronavirus spike sequences. *MDPI Biology*, 11(3):418, 2022. doi:10.3390/biology11030418.

32. Sarwan Ali, Yijing Zhou, and Murray Patterson. Efficient analysis of COVID-19 clinical data using machine learning models. *Medical & Biological Engineering & Computing*, 60(7):1881–1896, 2021. doi:10.1007/s11517-022-02570-8.

33. Zahra Tayebi, Sarwan Ali, and Murray Patterson. Robust representation and efficient feature selection allows for effective clustering of SARS-CoV-2 variants. *Algorithms*, 14(12):348, 2021. doi:10.3390/a14120348.

34. Andrew Melnyk, Fatemeh Mohebbi, Sergey Knyazev, Bikram Sahoo, Roya Hosseini, Pavel Skums, Alex Zelikovsky*, and Murray Patterson*. From alpha to zeta: Identifying variants and subtypes of SARS-CoV-2 via clustering. *Journal of Computational Biology*, 28(11):1113–1129, 2021. doi:10.1089/cmb.2021.0302.

35. Sarwan Ali, Simone Ciccolella, Lorenzo Lucarella, Gianluca Della Vedova, and Murray Patterson. Simpler and faster development of tumor phylogeny pipelines. *Journal of Computational Biology*, 28(11):1142–1155, 2021. doi:10.1089/cmb.2021.0271.

36. Simone Ciccolella*, Murray Patterson*, Paola Bonizzoni, and Gianluca Della Vedova. Effective clustering for single cell sequencing cancer data. *IEEE Journal of Biomedical and Health Informatics*, 25(11):4068–4078, 2021. doi:10.1109/JBHI.2021.3081380.

37. Simone Ciccolella, Mauricio Soto, Murray D. Patterson, Gianluca Della Vedova, Iman Hajirasouliha, and Paola Bonizzoni. gpps: An ILP-based approach for inferring cancer progression with mutation losses from single cell data. *BMC Bioinformatics*, 21(1):413, 2020. doi:10.1186/s12859-020-03736-7.

38. Simone Ciccolella, Camir Ricketts, Mauricio Soto Gomez, Murray Patterson, Dana Silverbush, Paola Bonizzoni, Iman Hajirasouliha, and Gianluca Della Vedova. Inferring cancer progression from single-cell sequencing while allowing mutation losses. *Bioinformatics*, 37(3):326–333, 2020. doi:10.1093/bioinformatics/btaa722.

39. Raffaella Rizzi, Stefano Beretta, Murray Patterson, Yuri Pirola, Marco Previtali, Gianluca Della Vedova, and Paola Bonizzoni. Overlap graphs and de Bruijn graphs: Data structures for de novo genome assembly in the big data era. *Quantitative Biology*, 7(4):278–292, 2019. doi:10.1007/s40484-019-0181-x.

40. Stefano Beretta*, Murray Patterson*, Simone Zaccaria, Gianluca Della Vedova, and Paola Bonizzoni. HapCHAT: Adaptive haplotype assembly for efficiently leveraging high coverage in long reads. *BMC Bioinformatics*, 19(1):252, 2018. doi:10.1186/s12859-018-2253-8.

41. Wandrille Duchemin, Yoann Anselmetti, Murray Patterson, Yann Ponty, Sèverine Bérard, Cedric Chauve, Celine Scornavacca, Vincent Daubin, and Eric Tannier. DeCoSTAR: Reconstructing the ancestral organization of genes or genomes using reconciled phylogenies. *Genome Biology and Evolution*, 9(5):1312–1319, 2017. doi:10.1093/gbe/evx069.

42. Andrea Bracciali, Marco Aldinucci, Murray Patterson, Tobias Marschall, Nadia Pisanti, Ivan Merelli, and Massimo Torquati. PWhatsHap: Efficient haplotyping for future generation sequencing. *BMC Bioinformatics*, 17(11):342, 2016. doi:10.1186/s12859-016-1170-y.

43. Murray Patterson*, Tobias Marschall*, Nadia Pisanti, Leo van Iersel, Leen Stougie, Gunnar W. Klau†, and Alexander Schoenhuth†. WhatsHap: Weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology*, 22(6):498–509, 2015. doi:10.1089/cmb.2014.0157.

44. Murray Patterson, Gergely Szöllősi, Vincent Daubin, and Eric Tannier. Lateral gene transfer, rearrangement, reconciliation. *BMC Bioinformatics*, 14(15):S4, 2013. doi:10.1186/1471-2105-14-S15-S4.

45. Mohammed El-Kebir*, Tobias Marschall*, Inken Wohlers*, Murray Patterson, Jaap Heringa, Alexander Schoenhuth, and Gunnar W. Klau. Mapping proteins in the presence of paralogs using units of coevolution. *BMC Bioinformatics*, 14(15):S18, 2013. doi:10.1186/1471-2105-14-S15-S18.

46. Ján Maňuch*, Murray Patterson*, Roland Wittler*, Cedric Chauve, and Eric Tannier. Linearization of ancestral multichromosomal genomes. *BMC Bioinformatics*, 13(19):S11, 2012. doi:10.1186/1471-2105-13-S19-S11.

47. Ján Maňuch*, Murray Patterson*, and Cedric Chauve. Hardness results on the gapped consecutive-ones property problem. *Discrete Applied Mathematics*, 160(18):2760–2768, 2012. doi:10.1016/j.dam.2012.03.019.

48. Roland Wittler, Ján Maňuch*, Murray Patterson*, and Jens Stoye. Consistency of sequence-based gene clusters. *Journal of Computational Biology*, 18(9):1023–1039, 2011. doi:10.1089/cmb.2011.0083.

49. Ján Maňuch* and Murray Patterson*. The complexity of the gapped consecutive-ones property problem for matrices of bounded maximum degree. *Journal of Computational Biology*, 18(9):1243–1253, 2011. doi:10.1089/cmb.2011.0128.

Conference Proceedings

50. Mansoor Ahmed, Yue-Tsz Fan, Hemanth Venkateswara, and Murray Patterson. Latent-kernel discrete diffusion for few-step generation. In *Beyond Next Token Prediction: Diffusion and Flow Models for Next-Generation Decoding (BeNTo Workshop of NeurIPS, Sydney, Australia, 2026)*. 2026. to appear. URL: `https://openreview.net/forum?id=iuTdDIaTrl`.

51. Brian Riccardi, Simone Moretti, Murray Patterson, Iman Hajirasouliha, Victoria Popic, Paola Bonizzoni, Simone Ciccolella, and Gianluca Della Vedova. Compactly representing a set of tumor phylogenies. In *the 14th International Conference on Computational Advances in Bio and Medical Sciences (ICCABS, Raleigh, NC, 2026)*. 2026. to appear.

52. Avais Jan, Qasim Zia, Naila Jan, and Murray Patterson. Risk-controlled large language models for selective medical question answering via two-stage conformal prediction. In *the 14th International Conference on Computational Advances in Bio and Medical Sciences (ICCABS, Raleigh, NC, 2026)*. 2026. to appear.

53. Manasa Mudunuri, Chandra Kiran Guntupalli, Shiraj Pokharel, Murray Patterson, and Mario Michael Kubek. Bounded conversational memory with hybrid retrieval and evidence highlighting for multi-session dialogue systems. In *the 27th Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL, Atlanta, Georgia, 2026)*, 786–794. 2026. URL: `https://aclanthology.org/2026.sigdial-1.55/`.

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